Creation and validation of the LEVANTE core tasks: Internationalized measures of learning and development for children ages 5-12 years

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Abstract

We present the Learning Variability Network Exchange (LEVANTE) core tasks, a set of nine short and engaging tasks designed to assess learning and development in children ages 5–12 years across a wide range of languages and cultures. Using a simple and uniform multi-alternative forced-choice format, these tasks measure constructs including math, executive function, language, reasoning, and social cognition and can be administered on a tablet or computer, both in person and remotely, with all materials openly available. We describe the design and selection of these tasks, and then report on their reliability and validity in a sample of 1034 children recruited from sites in Colombia, Germany, and Canada. Tasks are scored using multi-group item response theory models, allowing testing for measurement invariance. The parameters of these models can then be used to create computer adaptive versions of the tasks, allowing the entire battery to be given in around an hour. We discuss the use, ongoing refinement, and extension of these tasks in the service of creating an open dataset to describe variability in children's development and learning across contexts.

Keywords: cognitive development

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Introduction

Developmental variability and change in childhood is a topic of substantial theoretical and practical interest. Whether tracking children's growth over time, evaluating intervention outcomes, or examining environmental moderators, many scientific aims require accurate assessment tools (Rubio-Codina et al., 2016). Ideally, psychological measures are efficient, reliable, valid, and usable across ages, situations, and contexts. In practice, however, researchers face a landscape that falls far short of this ideal, with gaps stemming from limited age coverage, variability across contexts, and restricted access to high-quality measures (McHenry et al., 2023).

Because children's overall capacities are so dependent on their age, it can be very challenging to use the same measure across children of different ages (Cappa et al., 2021). Young children require simple tasks that are not verbally demanding, while older children can answer more complicated questions, and can become bored if given too many simple questions. In addition, younger children typically require shorter tasks, often reducing measurement reliability. Yet giving different tasks to different ages can mean that scores are not comparable, making tracking developmental growth challenging in many domains.

A second set of challenges concern cross-context comparisons. Ideal developmental measures should be validated in a global context and applicable to children across many cultures and languages (Bohn et al., 2024). Child development is an issue of global importance (Aboud & Yousafzai, 2015; Grantham-McGregor et al., 2007; Masten, 2014), yet the vast majority of developmental measures are developed in very specific (often English-speaking) contexts (Kidd & Garcia, 2022; Richter et al., 2019; Singh et al., 2023).

A final set of challenges has to do with accessibility. Many gold-standard measures are commercially distributed. They are costly for researchers to use, and in addition, publishers may place barriers on new translation and adaptation. Publishers also typically hold both item information and normative data closely, blocking many types of secondary investigation.

LEVANTE

The Learning Variability Network Exchange (LEVANTE) was designed to address these challenges (Frank et al., 2025) by validating an openly accessible set of measures applicable to a wide developmental age range within varied contexts. LEVANTE adopts a software framework from the Rapid Online Assessment of Reading (ROAR) to facilitate data collection with these measures. The ROAR dashboard is a web interface to assign both task bundles and surveys to groups of children, caregivers, and teachers. LEVANTE "forked" (duplicated) the ROAR dashboard and modified it for international research with the addition of a framework for data harmonization and validation. Data collected through the dashboard are harmonized into a centralized data repository and made accessible first to the researchers who collected them and then, through regular data releases, to the broader research community. Through a partnership with the Jacobs Foundation, sites around the world are funded to collect longitudinal data from children using the LEVANTE framework. The eventual goal of LEVANTE is to create and openly share a rich dataset and suite of measures documenting children's learning and development across contexts. Code and data for reproducing this paper are available at https://github.com/levante-framework/levante-pilots.

The current manuscript introduces the LEVANTE core tasks, a suite of behavioral measures for children, and draws on data from the now-completed pilot phase of the project, which we used to evaluate and refine the tasks' reliability, validity, and feasibility across sites and age groups. In our initial development of the task battery, we cast a broad net for important constructs in child development with well-accepted measures that had been used internationally. This process is described in Frank et al. (2025). The broad constructs that we selected were executive function, language, mathematics, reasoning, and social cognition, with these being instantiated through a number of well-accepted task types.

To create our core tasks, we selected pre-existing measures from the literature that tapped each of these constructs. When possible, we prioritized measures with strong psychometric properties, previous use across a broad range of cultures, applicability across a broad range of

ages, and lack of commercial or licensing constraints. Although very few measures met all our criteria, this search yielded a series of candidate measures, which we adapted and re-implemented in an open source web platform. Table 1 shows these tasks, organized by construct. Some of these implementations remained quite close to prior versions, while others, by necessity, required the creation of additional test items (e.g. for longitudinal retest and cultural appropriateness) or changes to specifics of their procedure.

In addition to the constructs described above, we were also interested in the assessment of literacy. The LEVANTE task battery makes use of a number of previously-validated literacy tasks from the Rapid Online Assessment of Reading (Yeatman et al., 2021), including Single Word Reading (ROAR-Word) (Ma et al., 2025a; Yeatman et al., 2021), Sentence Reading Efficiency (ROAR-Sentence) (Yeatman et al., 2024), and Phonological Awareness (ROAR-Phoneme) (Gijbels et al., 2024). Because they are extensively described elsewhere, we do not report on these tasks here, though we make use of literacy data for validation of language measures. For validation of our executive function measures, we also included a commercially-available broad measure of executive function, the Minnesota Executive Function Scale (MEFS), for which substantial reliability and validity evidence has already been gathered (Carlson, 2021).

Table 1

The LEVANTE core tasks, presented with their simplified labels as well as prior labels used in the literature.

Construct	LEVANTE task name	Prior/Source task name	Most similar source citation
Executive Function	Hearts and Flowers	AMES Hearts and Flowers	Davidson et al. (2006)
	Memory	AMES Memory Game	Corsi and Michael (1972)
	Same and Different	Something's the Same/Flexible Item Selection Task	Willoughby, Wirth, and Blair (2011)
			Jacques and Zelazo (2001)
Language	Vocabulary	Receptive Vocabulary	Dunn and Dunn (1965)
	Sentence Understanding	Test for Reception of Grammar (TROG)	Bishop (1982)
Math	Math	Early Grade Math Assessment (EGMA)	Platas et al. (2014)
Reasoning	Pattern Matching	Matrix Reasoning	Chierchia et al. (2019)
Spatial Cognition	Shape Rotation	Mental Rotation	Frick, Hansen, and Newcombe (2013)
Social Cognition	Stories	Theory of Mind	Sotomayor-Enriquez et al. (2024)

The current study

Here we report on the development and validation of the LEVANTE core tasks. This was an iterative process in which data from 5–12 year old children were collected at three sites: Bogota, Caquetá and Boyacá, Colombia (uniandes-co); Leipzig, Germany (mpieva-de); and Ottawa, Canada (western-ca).[1]

Our practices in this manuscript reflect an evolving set of measures – in this sense, LEVANTE is a "ship that is being built as we sail." In some cases, we used data gathered during the pilot data collection process to make minor changes to the tasks. We also used data from these three pilot sites to develop efficient, computer adaptive task (CAT) versions of many of the tasks. Thus, our report here reflects progress to date, rather than a single, monolithic process in which tasks were created and then piloted uniformly across all groups.

Psychometric models based on item-response theory (IRT) are a key component of our process for developing the LEVANTE core tasks (Embretson & Reise, 2013). IRT provides a family of models that allow the joint estimation of the difficulty of individual task items (e.g., math questions) and the ability of individual children. A fitted IRT model provides task parameters that can be used to estimate the ability of a new test taker given their responses on some or all of the same items. In addition, IRT parameters are used in the construction of CATs, which choose the most relevant subset of items to present, to estimate the ability of a particular individual. Critically for our purposes, the use of IRT models means that we can provide comparable scores on the same scale for both a younger child who saw mostly easier task items and an older child who saw harder items. These models thus allow us to address our first key challenge posed above: limited coverage across age groups.

Because our data come from three sites, each with their own translations and adaptations of the specific tasks, we can also use multi-group IRT models to explore the question of

[1] In the remainder of the manuscript, we refer to these sites by their site designators, uniandes-co, mpieva-de, and western-ca. These designators mark the research group and country of data collection. We make this choice to avoid the inference that these data are particularly representative of Canadian, German, or Colombian children more broadly.

measurement invariance: whether measures function similarly across different groups (Putnick & Bornstein, 2016). While measurement invariance is more commonly discussed in the factor analytic literature, it is also applicable to IRT (where it is sometimes analyzed at the level of individual test items as "differential item function" across groups) (Thissen, 2025). Here we use multi-group model comparisons (described below) to investigate whether our tasks operate similarly across groups. In particular, where possible, we aim for *scalar invariance*, in which different groups still show the same relative ordering of difficulty across items (e.g., fraction items are harder than division items in a math test). In some cases, we may fall back to *metric invariance*, in which items show different difficulties across groups (e.g., 7×9 is harder for one group than another), or *configural invariance*, in which items show different degrees of ability discrimination as well (e.g., if some problem types are unfamiliar to children in one group and so do not discriminate between high and low ability children). These models allow us to begin to address the second challenge posed above: variability across contexts.

Although we are interested in using psychometric models to create comparable scores for our tasks, the goal of LEVANTE is not to measure absolute differences between sites. Indeed, site comparisons are not possible in our framework. Our pilot sites and future LEVANTE sites differ widely in the details of sampling, recruitment, and administration, making differences in absolute performance across sites highly confounded. Thus, although below we present data that suggest absolute performance differences between sites, we caution that these are uninterpretable.

In what follows, we begin by describing the nine LEVANTE core tasks, organized by construct. We then discuss the process of translation and adaptation that produced the Spanish and German versions of these tasks from the original English source. Next we describe our pilot data collection. We subsequently present our IRT-based scoring techniques and the results of multi-group comparison, in some cases after selectively removing items that show poor psychometric properties. Using scores from these analyses, we then present evidence on the reliability and validity of the tasks, noting that many are still under development and that we expect their psychometric properties to improve as items are iteratively refined and more data are

collected.

We end by discussing future plans for further internationalization and downward extension of the tasks (i.e., adapting them for younger children). Critically, LEVANTE embraces open science values, aiming to create measures and data that are permissively licensed and reusable and extensible by the international research community. These values address our final challenge posed above: the aim of LEVANTE is to minimize barriers to reuse, accelerating progress towards a global science of learning and development.

The LEVANTE Core Tasks

General task properties

The LEVANTE core tasks are brief behavioral assessments that can be delivered in a web browser on a tablet or laptop, with responses made via touchscreen, keyboard, or mouse. Because administration formats vary, the tasks prioritize response accuracy rather than reaction time (and are therefore mostly untimed). However, reaction times are still recorded for all tasks. The tasks are deliberately designed for simplicity and clarity so they remain accessible to children across a wide age range. With only modest exceptions, nearly all follow a multi-alternative forced-choice format (up to four options) and use colored illustrations.[2] This uniformity of format means that in most cases, instructions can be short and easy to understand, minimizing delays when the tasks are given in sequence as a battery. Figure 1 shows screenshots from each of the tasks. Below we briefly present each task.

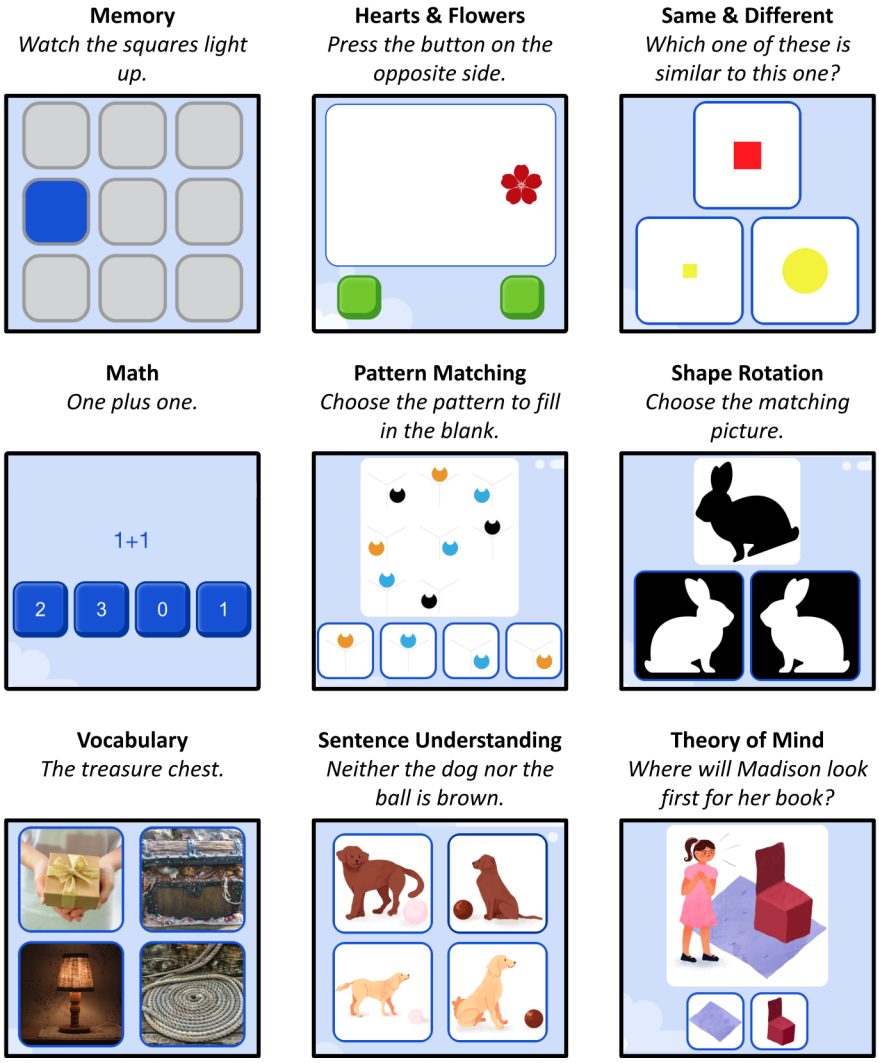
Tasks were implemented using jsPsych (De Leeuw, 2015) in a common framework. Illustrations were produced by hand, while auditory stimuli were generated using AI-based voice synthesis tools (https://elevenlabs.io), which enabled rapid and iterative revisions without the need for rerecording. The tasks generally share a set of usability features including simple instructions, a small number of practice trials, and buttons to replay audio.

All tasks are available for demonstration purposes at https://researcher.levante-network.org. Source code for the tasks is available at

[2] The vocabulary task uses photographs, and the math task shows only text.

Figure 1

Screenshots from each of the nine LEVANTE core tasks. Top row, from left to right: Sentence Understanding, Vocabulary, Hearts and Flowers; middle row: Memory, Same and Different, Math; bottom row: Pattern Matching, Mental Rotation; Stories.



https://github.com/levante-framework/core-tasks. Task code and assets are licensed CC-BY-NC 4.0 for non-commercial reuse (including educational use by not-for-profit and governmental entities) with appropriate attribution. Please see the license file available in the repository for more details. Once the researcher management dashboard is completed, interested researchers will be able to create accounts and use the tasks in their own research.

Table 2

Median task durations (minutes) for non-adaptive (left) and adaptive (right) versions of each task, within across all sites and within each site. Adaptive durations include in parentheses what percentage that adaptive duration is of the corresponding non-adaptive duration.

Task	Non-adaptive All	Non-adaptive mpieva-de	Non-adaptive uniandes-co	Non-adaptive western-ca	Adaptive All	Adaptive mpieva-de	Adaptive uniandes-co	Adaptive western-ca
Hearts & Flowers	2.94	3.14	2.89	2.59				
Sentence Understanding	7.60	8.88	6.79	7.01	3.76 (49%)	3.76 (42%)	3.77 (56%)	3.76 (54%)
Same & Different	9.69	9.71	4.32	9.81	5.74 (59%)			5.74 (59%)
Pattern Matching	3.93	9.66	1.18	5.19	5.38 (137%)		5.22 (442%)	5.41 (104%)
Shape Rotation	6.99	8.93	1.96	8.10	3.29 (47%)	3.29 (37%)	3.26 (166%)	3.32 (41%)
Stories	8.16	9.00	7.10	7.48				
Math	14.35	14.34	14.36	14.31	6.47 (45%)			6.47 (45%)
Memory	4.48	4.13	4.85	4.56				
Vocabulary	9.26	9.51	7.24	8.66	3.55 (38%)	3.47 (36%)	3.75 (52%)	3.63 (42%)

General task lengths are described below and average task durations from pilots (for both adaptive and non-adaptive versions) are given in Table 2. Because of our interest in testing children across a wide range of ages, we intentionally included trials that we anticipated would be both very easy and very hard for children with the aim of deploying these items adaptively after we had gathered sufficient data. We wanted to avoid frustration for children, however, so during pilot testing, we experimented with a number of different stopping rules. In early iterations of pilot testing at the Colombia site (uniandes-co), we ended tasks after three consecutive incorrect trials; we later modified this rule to end tasks after six consecutive incorrect trials. These differences explain some of the disparate task lengths that are observed (e.g., very short task lengths for Pattern Matching in uniandes-co).

Language

Sentence Understanding

The Test for Reception of Grammar (TROG) (Bishop, 1982) is a multiple-choice measure of receptive grammatical understanding. On each trial, the child hears a spoken sentence (e.g., "The cow is pushed by the girl.") and is asked to select one of four pictures that best matches its meaning. The original test contained 20 blocks, each with four items assessing the same grammatical structure. In our adaptation of the original TROG (which was permissively licensed for reuse), we removed several items that no longer reflected contemporary cultural norms, particularly those involving gender stereotypes. In addition, based on early pilot testing showing that many trials were easy for older children, we added a set of several dozen more challenging sentences. All illustrations were remade with details intended to be accessible across a broad range of cultures.

Vocabulary

The Vocabulary task was developed as a non-commercial, open alternative to tasks such as the Peabody Picture Vocabulary (Dunn & Dunn, 1965) and the NIH Toolbox Picture Vocabulary Task (Zelazo & Bauer, 2013); see Long et al. (2025) for more details. In this task, children hear a target word read aloud and are shown four pictures. They must identify the picture that matches

the word among three distractors. A first set of targets and distractors were sourced from the THINGS+ dataset (Hebart et al., 2019; Stoinski et al., 2024), specifically from the permissively-licensed subset of the data; in this item subset (108 items) each image had a semantically close and semantically far distractor as well as an unrelated distractor image (e.g., target word “acorn” with close distractor being a “coconut”, and the far and unrelated distractors being “keys” and “laundry”). Additional difficult items were included beyond those analyzed in Long et al. (2025). These more difficult items were sourced and translated from an open-sourced Dutch receptive vocabulary assessment tool, the DAIVT (Bousard & Brysbaert, 2021); we also manually constructed several of our own items to reach a total of 170.

Math

We developed the Math task based on the EGMA (Early Grade Mathematics Assessment) (Platas et al., 2014). This short, paper-and-pencil assessment is widely administered in international contexts for children ages 5–8 and includes number identification, number comparison, missing number, addition, and subtraction sub-tests. In our adaptation, we increased the breadth of the initial item bank, added multiplication, division, and fractions items to extend up to content that might be encountered by approximately age 12 years. We also included number line identification problems in which children were asked to place a marker on a number line across a range of scales (including simple 0–10 trials, larger whole-number scales, and a fraction scale). Number line tasks of this type are known to be strongly associated with mathematical ability (Schneider et al., 2018).

Reasoning

Pattern Matching

In classic matrix reasoning tasks, participants see a set of abstract figures in a grid format with a missing cell and must choose the most appropriate figure to complete the grid. Matrix reasoning scores are highly correlated with general cognitive ability and educational attainment (e.g., Roth et al., 2015). Here we make use of the Mars-IB matrix reasoning stimuli, an open-source, permissively-licensed set of matrix reasoning problems with retest variants that have

been normed with a large sample of adolescents and adults (Chierchia et al., 2019).

Shape Rotation

Mental rotation refers to the ability to imagine the rotation of objects in space, which is strongly related to mathematics and reasoning abilities (Xie et al., 2020). In our version of the mental rotation task (based on Frick et al., 2013), children are presented with a target item (e.g., a picture of a rabbit). Then, they are presented with two comparison items (e.g., two silhouettes of rabbits). One of the comparison items will match the target item when rotated; the other comparison item will not match the target item when rotated. The child must select the comparison item that matches the target item. Both accuracy and reaction time are measured. Across trials, the degree of rotation varies (e.g., 40, 80, 120, 160 degrees), affecting children's accuracy and reaction time. We included both two-dimensional duck and rabbit stimuli as well as harder three-dimensional geometric figures from the original stimulus set developed by Shepard and Metzler (1971).

Executive Function

Same and Different

The Same Different Selection (SDS) task was designed to assess cognitive flexibility in children (Obradović et al., 2025). It incorporates elements from the "Something's the Same" task (Willoughby et al., 2011) and "Flexible Item Selection Task" (Jacques & Zelazo, 2001) as initial task blocks and extends these tasks by including more advanced blocks that increase the number of conceptual dimensions, values per dimension, and number of presented cards. Children are presented with sets of cards that show geometric figures that vary along multiple dimensions, such as shape, color, size, and number. They are required to identify similarities and differences between cards based on these dimensions (e.g., choosing a pair of cards with figures that share the same color and then choosing a pair of cards that share the same number of figures), by engaging their ability to shift attention from previously identified dimensions and identify new matching dimensions criteria. The version of the task we adapted included successively more difficult blocks of trials in which children were asked to identify two, three, or four pairs of items that were

the same in different ways (e.g., matching on different dimensions).

Hearts and Flowers

The Hearts and Flowers task assesses inhibitory control and cognitive flexibility (Davidson et al., 2006). We modeled the task design after the AMES tablet task version which has been tested in many global settings including Ghana and Côte d'Ivoire (Ahmed et al., 2022; Khan et al., 2024). Participants respond according to stimulus type: pressing a key on the same side as a heart (congruent rule), and on the opposite side for a flower (incongruent rule). The task includes three blocks: congruent (hearts only), incongruent (flowers only), and mixed (hearts and flowers). The congruent block serves as a baseline with minimal executive demands. Participants in our version of the task are encouraged to respond as quickly as possible but – due to variation in administration platform and in contrast to some versions of the task – we did not provide a hard time limit on each trial, though we scored responses slower than 3000 ms as incorrect.

Memory

The Corsi Block task is a widely-used measure of visuospatial short-term and working memory (Corsi & Michael, 1972). In our version of the task modeled after the AMES tablet task version (Obradović & Sulik, 2025), the child sees a grid of squares; during each trial, a subset of squares lights up one at a time in a specific sequence. The child is required to reproduce the sequence by touching the squares: in the first block, the sequence is reproduced in the same order and in the second block, it is reproduced backwards. The task begins with short sequences (e.g., two items) and gradually increases in difficulty (up to seven items) until the child fails two sequences of the same length. We used both 2x2 and 3x3 grids in our pilot testing, with relatively similar results.

Social Cognition

Stories

Assessing social cognition across a wide range of ages is a challenge, and relatively few instruments show good reliability and validity across not just early childhood but older children as

well (cf. Heise & Bowman, 2025). In addition, due to the lengthy nature of vignette-based assessments, it was challenging to identify an assessment that would fit in our allotted time. We chose to adapt and re-illustrate a storybook task developed by Sotomayor-Enriquez et al. (2024) in which children hear stories with multiple social reasoning questions embedded, including questions about true and false beliefs, deception, and moral reasoning. Drawing on published data from this task, we selected items with good psychometric properties and strong potential for cross-cultural validity (e.g., stories commonly found across many cultures), and then supplemented these with additional trials assessing emotional reasoning. The result was a set of six stories, each with approximately five forced-choice social reasoning questions embedded within each story. We then developed two sets of alternative retest stories, in each case using the same structure as an existing story but substituting new details and characters.

For each task we distinguish between trials and IRT items (see Table 3). Trials refer to individual questions that could be presented to the child (design-level count; e.g., 60 trials in the non-adaptive Hearts & Flowers task). For psychometric analyses, we define a smaller set of IRT items, where repeated trials of the same type are collapsed into a single scored unit and practice or feedback trials are excluded. These IRT items are the units that enter the IRT models and therefore have estimated discrimination and difficulty parameters.

Table 3

Number of trials and IRT items per LEVANTE task.

Task	Total trials (questions)	IRT items (analytic units)
Hearts and Flowers	60	10
Memory	41	27
Same and Different	127	15
Vocabulary	170	161
Sentence Understanding	99	98
Math	202	359
Pattern Matching	78	156
Shape Rotation	81	12
Stories	6	89

Note. "Total trials" gives the number of distinct questions in the non-adaptive version of each task. "IRT items" gives the number of scored units included in IRT modelling after collapsing repeated trials, excluding practice/feedback trials, and other item exclusions (see Scoring section).

Translation and adaptation

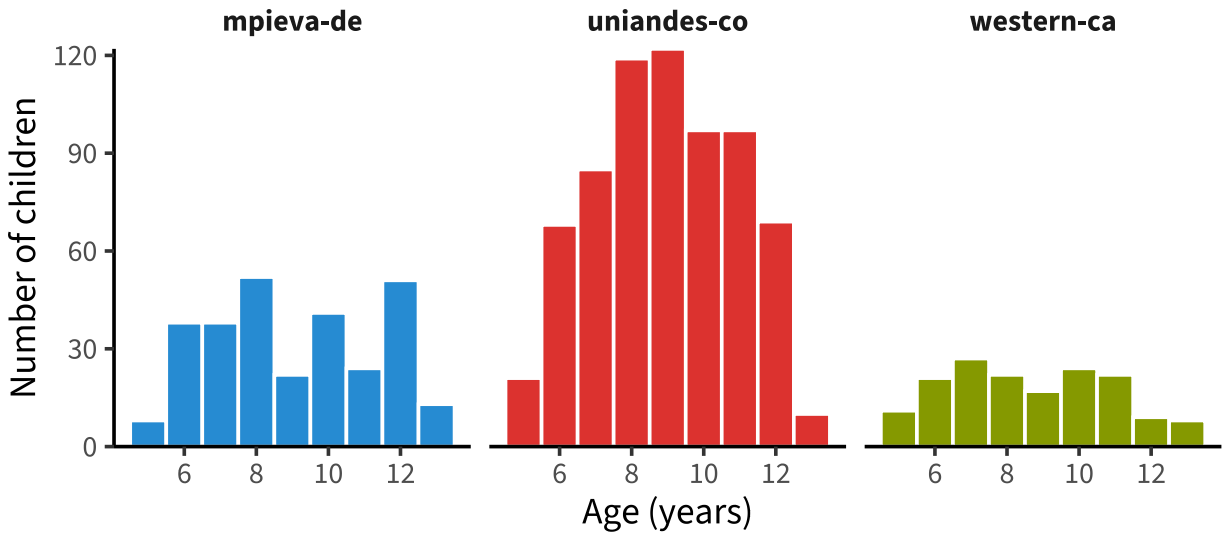
Initially designed in English, the core tasks were subsequently adapted for Spanish and German following a set of internationalization procedures in collaboration with researchers at the uniandes-co and mpieva-de sites, respectively. We began with AI-generated translations of all verbal materials, including task instructions, items, and other key terminology such as encouragement phrases that appear throughout the core tasks; we used the DeepL platform for these initial translations. A professional translator then reviewed and edited the AI translations for general grammatical correctness and appropriateness. Materials were next reviewed by a researcher who was a native speaker of the relevant language and dialect, confirming that the materials were clear and appropriate for the specific task. We then conducted back translation using AI translation tools to verify semantic and conceptual equivalence and to flag items needing further review by native speakers. For tasks evaluating language skills (Vocabulary and Sentence Understanding), we additionally recruited PhD-level linguists with expertise in language acquisition for both Spanish and German to review items and make recommendations about relevant changes or additions to the item set for that language. Additional language adaptation work for these two tasks is ongoing based on the results of our pilot data collection.

Pilot Data Collection

The pilot site partnership was an integral component of the core task development. Across three countries and languages, pilot sites provided collaboration on task adaptation and functionality, iterative task testing, and diverse settings for infrastructure deployment. Data collected from pilot sites allowed analyses of construct validity, measurement invariance, test-retest reliability, parameters for computer adaptive testing, and a demonstration of age-related differences measured by the tasks across constructs. Because we continued developing tasks continuously during the ~18-month pilot process, we anticipate that our reported reliability and validity estimates are an underestimate of the current task performance. Our analyses pool over early versions of the tasks that in some cases had instructions or items that were later revised for clarity.

A key feature of the LEVANTE framework is that all data are fully de-identified, minimizing legal and ethical barriers to data sharing. No sociodemographic information or other identifying details about the child are entered into the researcher dashboard by participating sites, and ages are recorded only to the nearest month (Frank et al., 2025). We anticipate that sites will add sociodemographic information at a later stage via an app currently under development, which is designed to ensure that individuals cannot be statistically re-identified (e.g., due to rare combinations of characteristics within a community). For this reason, here we provide only minimal demographic characterization and analysis of the children from our pilot sites. Figure 2 provides the distribution of ages for each site.

Figure 2
Overall distributions of ages for each site.



uniandes-co

The Universidad de Los Andes team conducted school-based data collection at four schools in Bogotá and two public schools in the rural departments of Caquetá and Boyacá, selected for convenience from schools that had participated in previous projects with the university. In Bogotá, three of the participating schools belong to two family compensation funds and the fourth is managed by a charter-school organisation; across all sites, the schools primarily

serve children from lower-middle and low-income families. By partnering with these schools, the research team oversaw data collection using a group testing format on tablets provided by the team, with larger groups for older children and smaller groups for younger children. The uniandes-co site collected data from 1222 children ages 5–12 years. Due to time constraints for school-based administration, children typically completed a subset of the tasks.

mpieva-de

Data for the mpieva-de site were collected by the Max Planck Institute for Evolutionary Anthropology using family-based remote assessment. Families were recruited through an existing database of Leipzig residents and were first contacted by telephone to invite participation. A total of 292 children, aged 5–12 years participated. Families who agreed to partake received login instructions to complete the assigned LEVANTE tasks at home using their own computer or tablet. Mpieva-de also provided retest data, with follow-up assessments administered at either a 3-month or 7-month interval after the inital session. Participants were drawn from Leipzig, a medium-sized urban center of approximately 620,000 residents that is culturally diverse, with around 17% of the population having a migration background (primarily from Syria, Ukraine, and the Russian Federation). The population is highly educated, with most adults completing at least nine years of formal schooling and many holding high school and university degrees. Most children in Leipzig attend subsidized external childcare from infancy through primary school.

western-ca

Partners at Western University (Ontario) recruited participants from the local community, primarily through online outreach (Facebook advertisements and community group posts), supplemented by posters on neighborhood mailboxes and notices shared via local school newsletters. Interested families voluntarily registered to be included in our participant database. Children were tested individually by a researcher in a dedicated testing space on campus. The western-ca site collected data from 171 children, ages 5-12 years. Families were primarily recruited from London, Ontario, with a smaller subset from the Greater Toronto Area (GTA), both predominantly urban regions. According to recent census estimates, London residents have a

mean education level of some postsecondary education, and the median household income is approximately $80,000 CAD.

Data filtering

We use “run” to represent an instance of a child attempting a single task. Across all tasks, the sites collected a total of 9479 runs (mpieva-de: 4529, uniandes-co: 3458, western-ca: 1492) from 1685 unique children (mpieva-de: 292, uniandes-co: 1222, western-ca: 171), of which some runs were excluded for not having any associated trial data (mpieva-de: 369 [8.1%], uniandes-co: 121 [3.5%], western-ca: 131 [8.8%]); for being incomplete, invalid, or duplicate (mpieva-de: 447 [9.9%], uniandes-co: 492 [14.2%], western-ca: 180 [12.1%]); for missing item metadata (mpieva-de: 0 [0.0%], uniandes-co: 107 [3.1%], western-ca: 0 [0.0%]); for missing age (mpieva-de: 0 [0.0%], uniandes-co: 599 [17.3%], western-ca: 0 [0.0%]); for for age being out of range (mpieva-de: 123 [2.7%], uniandes-co: 28 [0.8%], western-ca: 21 [1.4%]), or for having too short (<0.6 months) of a retest gap (mpieva-de: 23 [0.5%], uniandes-co: 0 [0.0%], western-ca: 16 [1.1%]), resulting in a total of 6822 scored runs (mpieva-de: 3567, uniandes-co: 2111, western-ca: 1144) from 1136 unique children (mpieva-de: 287, uniandes-co: 688, western-ca: 161).

Within the included runs, there were a total of 413517 trials (mpieva-de: 204346, uniandes-co: 152290, western-ca: 56881), of which 5910 [1.4%] were excluded (mpieva-de: 3188 [1.6%], uniandes-co: 2080 [1.4%], western-ca: 642 [1.1%]) due to too fast (<100ms) or too slow (>30s) reaction times. For reasons of space we do not disaggregate these exclusions by task, but note that the task with the highest exclusion rate was Pattern Matching (9.3% trials excluded), while all the other tasks ranged between 0.1% and 2.2% trials excluded.

Scoring

Our goal for the psychometric modelling of the data obtained in each task was to extract administration-level ability scores (i.e., at the level of a testing session) with the highest possible reliability and validity. To do so, we first conducted heuristic item screening to remove poorly functioning items, then fit a sequence of multi-group IRT models to the retained items to estimate latent ability scores. These analyses provided the foundation for the reliability, developmental,

and validity results described in subsequent sections.

Heuristic item pruning and data checking

Prior to fitting psychometric models, we examined item-level performance summaries for each task within each site to identify items that were functioning unusually poorly (e.g., showing average accuracy near or below chance). We also reviewed classical diagnostics such as proportion correct, item–total correlations, and missingness rates to flag items that showed little discrimination or appeared anomalous. In some cases this was due to potential translation issues; in others it reflected aspects of item design. We manually excluded these items from further processing (N = 35; 23 items in Stories, 1–4 items in each of Same and Different, Math, Pattern Matching, Sentence Understanding, and Vocabulary). In one case, this process enabled us to identify a scoring bug affecting three tasks. We excluded the affected observations, which were collected during a two-day testing period at the mpieva-de site. The resulting cleaned calibration dataset formed the basis for all subsequent IRT estimation and invariance analyses.

Multi-group IRT model selection

Following item screening, we evaluated measurement invariance across sites using nested multi-group IRT models to assess the comparability of latent ability estimates. The conventional hierarchy of factor-analytic invariance—configural → metric → scalar—was implemented in the IRT framework as successive equality constraints on item parameters (see Table 4 for correspondence and interpretation). Configural invariance reflects a shared latent structure, metric invariance constrains item discriminations, and scalar invariance constrains both discriminations and difficulties across sites. For each task, we compared unidimensional Rasch and two-parameter logistic (2PL) models under these nested invariance constraints.

All models were estimated using the `mirt` R package (Chalmers, 2012) via marginal maximum likelihood using 500 quadrature points, with expected a posteriori (EAP) scoring. Overlapping items across languages served as anchors to establish a common scale; non-overlapping items were freely estimated within the same multi-group calibration. Model convergence and identification were verified by ensuring stable log-likelihoods and Hessian

condition numbers below 10^3. We specified Gaussian priors on item difficulty (mean = 0, SD = 3) and discrimination (mean = 1, SD = 0.3) parameters in all models.

Model selection was based on the Bayesian Information Criterion (BIC), with the final model for each task representing the most parsimonious combination of model type (Rasch, 2PL) and invariance constraints that adequately reproduced cross-site response patterns. Across domains, Rasch models generally provided the best fit, whereas 2PL models were preferred for Same and Different and Shape Rotation, which showed greater variability of discrimination parameters across items.

As shown in Table 6, seven tasks – Hearts and Flowers, Same and Different, Sentence Understanding, Pattern Matching, Shape Rotation, Math, and Memory – reached scalar invariance, indicating that both discrimination and difficulty parameters were comparable across sites. Two tasks – Stories and Vocabulary – showed configural invariance, suggesting a shared latent construct with certain parameters differing across groups.

Overall, these results suggest that most tasks showed good measurement equivalence across sites. The final multi-group calibrations provided a common metric for subsequent ability estimation.

Score computation

Individual ability scores θ were estimated from the final multi-group calibrations using the expected-a-posteriori (EAP) method. Scores are expressed on a common latent scale across sites, allowing for direct cross-site comparisons and forming the basis for the developmental sensitivity, reliability, and validity analyses that follow.

Table 4

Measurement invariance across factor analysis and item response theory.

Invariance	Model	mirt invariance param	Factor-analytic analog	IRT explanation	Practical interpretation
None	–	–	Exploratory factor analysis	Not possible because IRT assumes unidimensionality	Groups differ in meaning of construct
Configural	Rasch	“”	–	Non-uniform DIF: Groups differ in all item characteristics	Groups express construct in different ways
Configural	2PL	“”	Configural: Same # of factors, loadings = 0 for same items	Non-uniform DIF: Groups differ in all item characteristics	Groups express construct in different ways
Metric	2PL	`c(“slopes”)`	Metric (weak): Equal item factor loadings	Uniform DIF: Equal item discriminations (a = a_g)	Groups express construct in similar ways
Scalar	Rasch	`c(“intercepts”, “free_means”, “free_vars”)`	–	No DIF: Equal item difficulties (d = d_g)	Factor scores are on same scale and can be compared across groups
Scalar	2PL	`c(“slopes”, “intercepts”, “free_means”, “free_vars”)`	Scalar (strong): Equal item factor loadings & item intercepts	No DIF: Equal item discriminations & item difficulties (a = a_g, d = d_g)	Factor scores are on same scale and can be compared across groups
Full	–	–	Full (strict): Equal item factor loadings, intercepts, residual variances	Not possible because there are no residual variances with binary items in IRT	Factor scores are on same scale and can be compared across groups; Reliabilities also equivalent

Psychometric properties of tasks

Developmental sensitivity

Our goal for the LEVANTE core tasks is that they are sensitive to developmental change. In cross-sectional data, the age-gradients provide an approximation of how sensitive a task is to developmental differences and may, when longitudinal data are available, capture within-person change. We reiterate here that we do not believe absolute differences between sites are interpretable, because of confounding due to differences in sampling, recruitment, and administration of the tasks; our interest instead is in within-site developmental change.

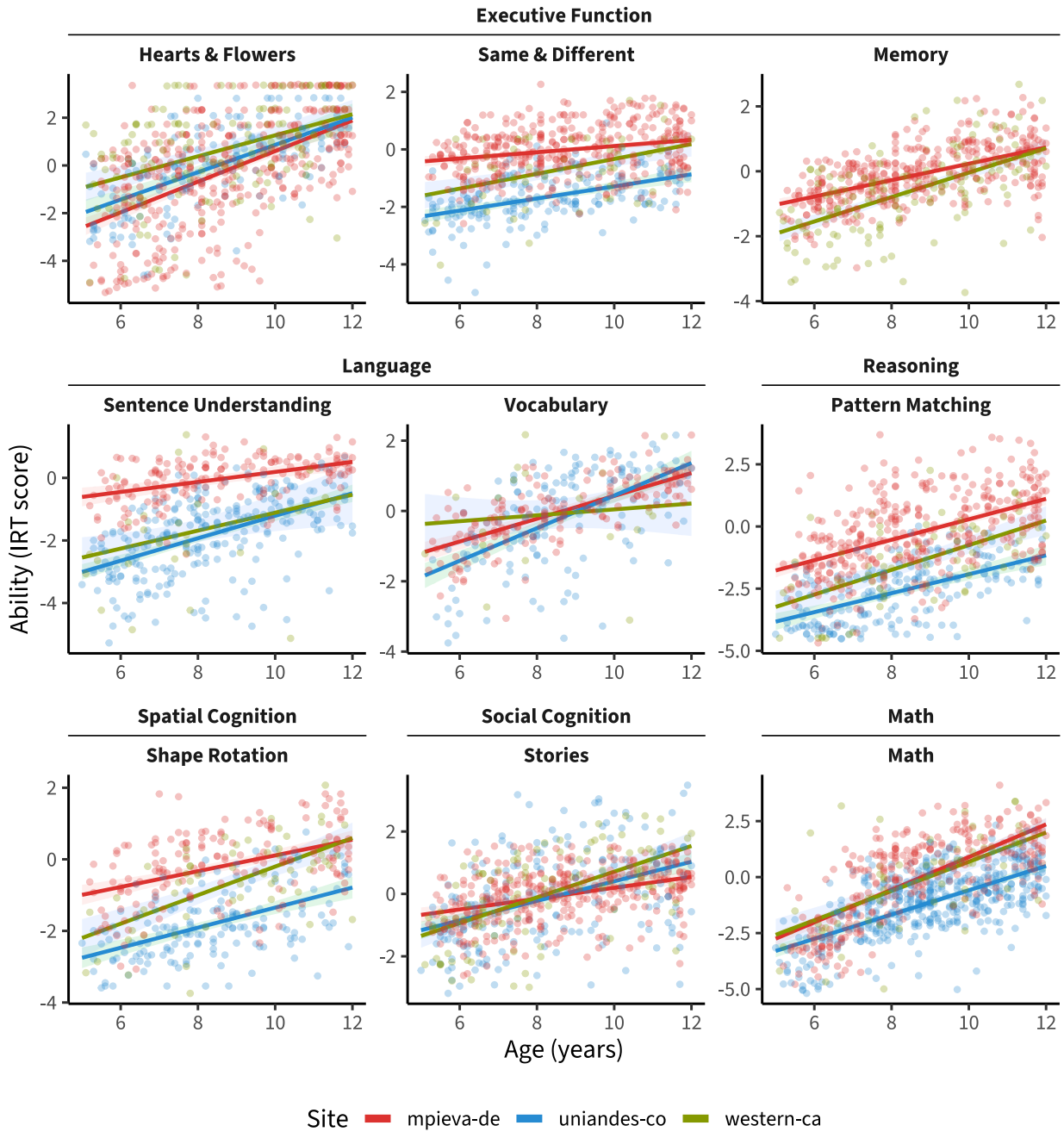
Across all nine tasks, ability scores were higher at older ages at all sites, indicating robust positive age gradients (see Figure 3). This pattern was particularly pronounced for Vocabulary, Hearts and Flowers, and Math. Age gradients were broadly similar across sites: the panels suggest parallel developmental gradients in cognitive performance, with systematic shifts in mean ability but little evidence that the age-related slope differs markedly by site.

To quantify the age gradients, we fit separate ordinary least-squares models predicting ability scores (IRT units) from centered age with site fixed effects and an age × site interaction, allowing slopes to vary by site. As summarized in Table 5, age-related slopes were positive for all tasks and sites, indicating that older children performed better on every measure (see Figure A1 in the appendices for visual representation).

For ease of interpretation, we calculated the zero-order correlations between age and IRT ability scores for each site. As seen in the right side of Table 4, these correlations were uniformly positive (r = .40 to .70), of moderate magnitude across tasks, and broadly similar across sites. In sum, the results of the age-based analyses suggest that the tasks are sensitive to age-related differences and are likely suitable for deployment in longitudinal studies of developmental change.

Figure 3

Estimated IRT ability by age across tasks and sites, non-adaptive data only.



Note. Uniandes-co Memory data were excluded due to task bugs and difficulties with the instructions.

Table 5

Age-Related Change in Performance: Task Grand Means (GM), Site Deviations (Δ), and Age–Ability Correlations (r), non-adaptive data only. Note: for each task, z-scores were regressed on centred age and site, with sum-to-zero contrasts for site. GM β is the grand-mean age slope, Δβ is each site's deviation from this slope, site-specific slope = GM β + Δβ. r is the within-site Pearson correlation between age and task score.

Task	Site	GM β	GM SE	GM p	Δβ	Δβ SE	Δβ p	N	r	r p
Hearts & Flowers	mpieva-de	0.55	0.04	<0.001	0.09	0.05	0.090	384	0.55	<0.001
Hearts & Flowers	uniandes-co	0.55	0.04	<0.001	0.02	0.06	0.703	164	0.66	<0.001
Hearts & Flowers	western-ca	0.55	0.04	<0.001	-0.11	0.06	0.090	137	0.42	<0.001
Sentence Understanding	mpieva-de	0.27	0.04	<0.001	-0.11	0.04	0.041	167	0.46	<0.001
Sentence Understanding	uniandes-co	0.27	0.04	<0.001	0.09	0.04	0.054	244	0.53	<0.001
Sentence Understanding	western-ca	0.27	0.04	<0.001	0.02	0.07	0.796	27	0.31	0.117
Same & Different	mpieva-de	0.19	0.02	<0.001	-0.08	0.03	0.005	365	0.24	<0.001
Same & Different	uniandes-co	0.19	0.02	<0.001	0.02	0.03	0.545	183	0.48	<0.001
Same & Different	western-ca	0.19	0.02	<0.001	0.07	0.04	0.161	59	0.42	<0.001
Pattern Matching	mpieva-de	0.43	0.03	<0.001	-0.02	0.04	0.634	353	0.51	<0.001
Pattern Matching	uniandes-co	0.43	0.03	<0.001	-0.05	0.04	0.389	203	0.58	<0.001
Pattern Matching	western-ca	0.43	0.03	<0.001	0.07	0.06	0.389	67	0.60	<0.001
Shape Rotation	mpieva-de	0.30	0.03	<0.001	-0.08	0.03	0.041	162	0.45	<0.001
Shape Rotation	uniandes-co	0.30	0.03	<0.001	-0.02	0.03	0.572	166	0.52	<0.001
Shape Rotation	western-ca	0.30	0.03	<0.001	0.10	0.04	0.041	59	0.62	<0.001
Stories	mpieva-de	0.30	0.02	<0.001	-0.13	0.03	<0.001	349	0.39	<0.001
Stories	uniandes-co	0.30	0.02	<0.001	0.01	0.03	0.663	199	0.43	<0.001
Stories	western-ca	0.30	0.02	<0.001	0.11	0.04	0.005	103	0.58	<0.001
Math	mpieva-de	0.64	0.03	<0.001	0.09	0.04	0.040	298	0.72	<0.001
Math	uniandes-co	0.64	0.03	<0.001	-0.10	0.04	0.029	461	0.65	<0.001
Math	western-ca	0.64	0.03	<0.001	0.01	0.06	0.848	59	0.61	<0.001
Memory	mpieva-de	0.31	0.02	<0.001	-0.06	0.02	0.003	384	0.57	<0.001
Memory	western-ca	0.31	0.02	<0.001	0.06	0.02	0.003	143	0.57	<0.001
Vocabulary	mpieva-de	0.29	0.04	<0.001	0.03	0.05	0.481	169	0.71	<0.001
Vocabulary	uniandes-co	0.29	0.04	<0.001	0.17	0.05	0.002	143	0.57	<0.001
Vocabulary	western-ca	0.29	0.04	<0.001	-0.21	0.08	0.014	19	0.11	0.646

Reliability

One goal for the LEVANTE core tasks is strong reliability both within an individual administration (across test items) and across multiple administrations separated in time. We report on each of these.

Marginal reliability

Marginal reliability, an IRT-based analogue to Cronbach's α, quantifies the proportion of true-score variance captured by a test based on the item information function (Thissen & Wainer, 2001). Unlike classical coefficients, which assume uniform measurement precision across the ability range, marginal reliability incorporates the fact that precision varies across θ and estimates overall reliability by integrating information across the assumed population distribution (Embretson & Reise, 2013). In multi-group IRT models, marginal reliability is computed separately for each site-specific sub-model. Table 6 reports marginal reliabilities for each task and site from the best-fitting model. Reliabilities were largely acceptable (>.60) (Nunnally, 1978), with the exception of the Stories and Hearts and Flowers tasks, which showed lower reliability in several sites.

Table 6
Marginal reliabilities for each task and site.

Task	Model	Invariance	Reliability mpieva-de	Reliability uniandes-co	Reliability western-ca
Hearts & Flowers	Rasch	Scalar	0.54	0.66	0.59
Sentence Understanding	Rasch	Scalar	0.87	0.71	0.65
Same & Different	2PL	Scalar	0.91	0.91	0.85
Pattern Matching	Rasch	Scalar	0.68	0.75	0.67
Shape Rotation	2PL	Scalar	0.88	0.85	0.83
Stories	Rasch	Configural	0.77	0.52	0.45
Math	Rasch	Scalar	0.87	0.91	0.86
Memory	Rasch	Scalar	0.96	0.90	0.93
Vocabulary	Rasch	Configural	0.90	0.84	0.83

Note. Cells in blue indicate reliability < .60.

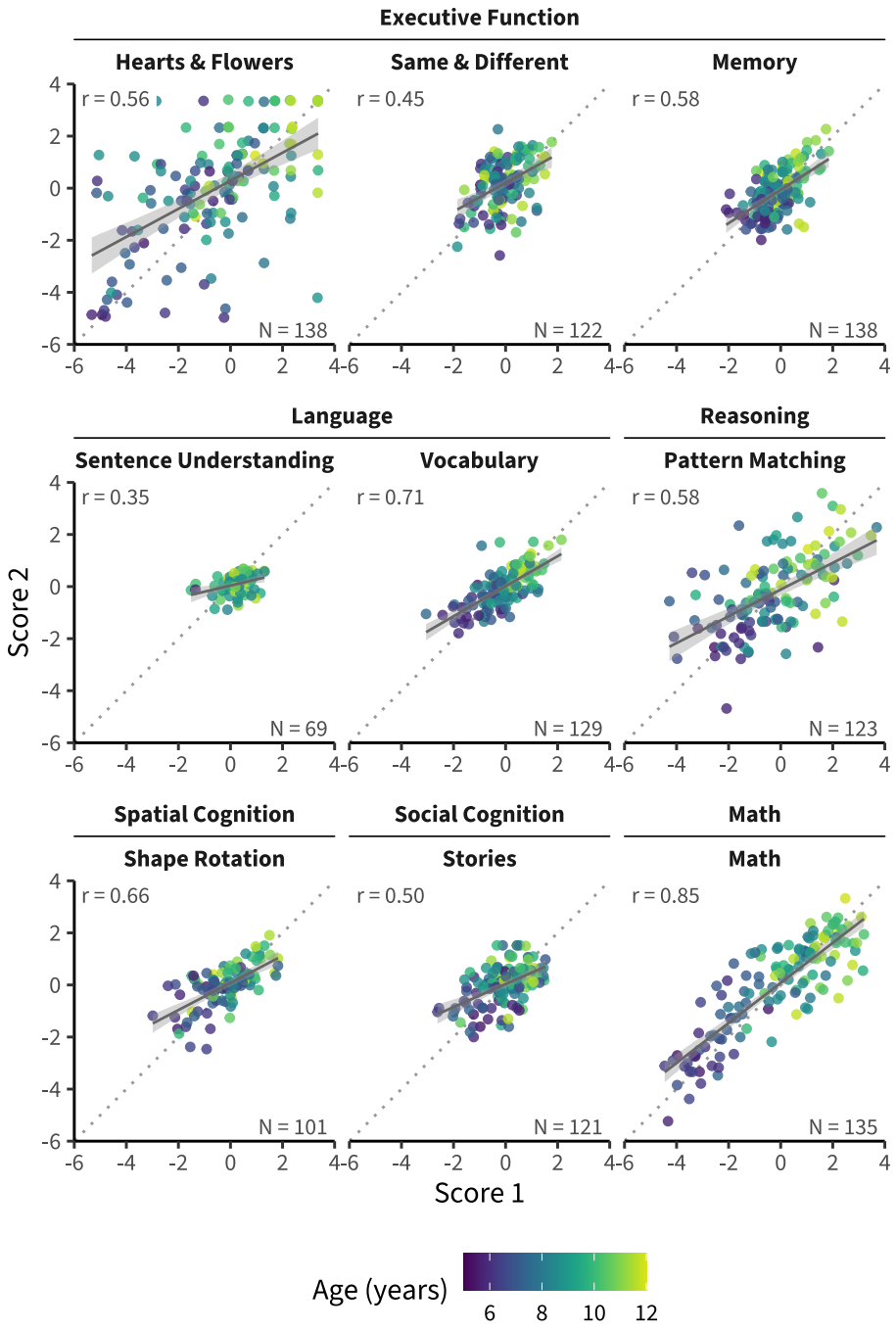
Test-retest reliability

We were also interested in capturing the extent to which the LEVANTE tasks yield reliable signal across a delay. The mpieva-de pilot site was able to re-administer tasks to a subset of their original participant group with a retest interval of either 3 months (group 1) or 7 months (group 2) approximately. We used this re-administration as an opportunity to collect data with computer adaptive versions of many of the tasks (Sentence Understanding, Pattern Matching, Shape Rotation, Math, and Vocabulary); see below for further details of computer adaptive versions. In addition, for several tasks (Math, Stories, Pattern Matching), we used this opportunity to gather item data about new groups of retest items that were designed to match the content of the original items but vary in their surface forms. For the other tasks, we retested using the same items.

Figure 4 shows the relations between time 1 and time 2 scores. Across individual tasks, retest sample sizes ranged from N = 69 to N = 138; the low number of repeat administrations for Sentence Understanding reflects an error in which this task was not sent to younger children. Children in the test–retest subsample were very similar in age to those in the overall mpieva-de pilot sample. The overall sample (n = 252) had a mean age of 8.77 years (SD = 1.96), whereas the test–retest subsample (n = 148) had a mean age of 8.44 years (SD = 1.87), a difference of approximately -0.32 years (~-4 months). Each figure panel also shows the test-retest correlation and sample size for the corresponding task. Most retest correlations were in the range of r = .5 – .7, with two exceptions. Sentence Understanding had a smaller sample size with a more restricted age range and showed lower reliability within that range; and Math showed higher reliability than the other tasks. Under classical test theory assumptions, these test–retest correlations provide an estimate of score reliability that complements the marginal reliabilities reported above (Kline, 2013).

Figure 4

Test-retest correlations per task (in mpieva-de data).



Validity

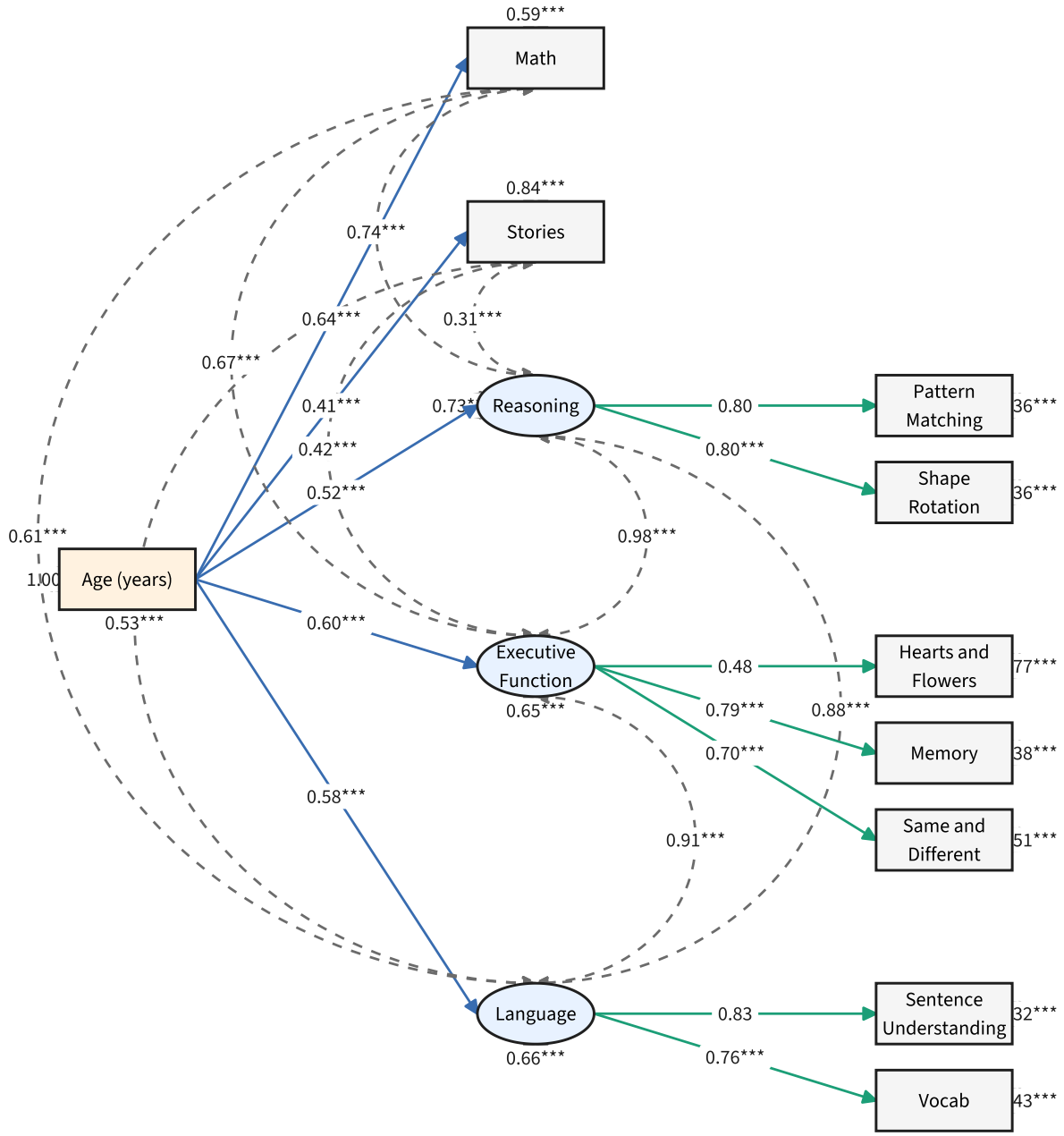
Assessment of validity for the core tasks was challenging, in part because of the precise issues that motivated us to create the task battery in the first place. No readily available, internationalized measures existed to compare to many of our tasks. Thus, for validation evidence, we rely on the overall construct structure of the core task battery as well as the relations to our gold standard measures for literacy (ROAR-Word, ROAR-Sentence and ROAR-Phoneme) and executive function (MEFS).

Construct validity

We assessed construct validity using a structural equation model (SEM) specifying three first-order latent factors—Reasoning, Executive Function, and Language—with each factor measured by its respective tasks and allowed to covary with Math and Stories (see Figure 5). Math and Stories were treated as observed endogenous outcomes because each construct had a single task; introducing one-indicator latent variables would require strong, unverifiable assumptions about reliability and error variance. Age was specified to predict each of the latent factors as well as the observed variables Math and Stories.

Figure 5

Structural Equation Model for core tasks. Note: Ovals represent latent factors; rectangles represent observed variables. Blue fill indicates latent factors, orange indicates the age predictor, and grey indicates observed indicators and outcomes. Green arrows represent factor loadings; blue arrows represent regression paths; grey dashed double-headed arrows represent residual variances and covariances (error terms). Values on paths are standardised coefficients.



Overall model fit did not meet conventional thresholds, CFI: 0.927, RMSEA: 0.093 RMSEA p-value: <.001 (Hu & Bentler, 1999). Factor loadings supported the intended three-factor structure (convergent validity), and age showed positive associations with each domain. The high inter-factor correlations indicate that much of the variance is shared across domains.

Comparison with more parsimonious alternatives—including a single-factor model, a two-factor model separating reasoning and language, and a three-factor model distinguishing reasoning, executive function, and language—showed that none provided better fit than the original model that treated Math and Stories as distinct outcomes (see Table 7). We therefore retain the three-factor SEM incorporating age effects and treating Math and Stories as observed outcomes, on both conceptual and empirical grounds. Future work will pursue more complete models that examine if and how the constructs may be organized differently across different contexts and ages.

Table 7

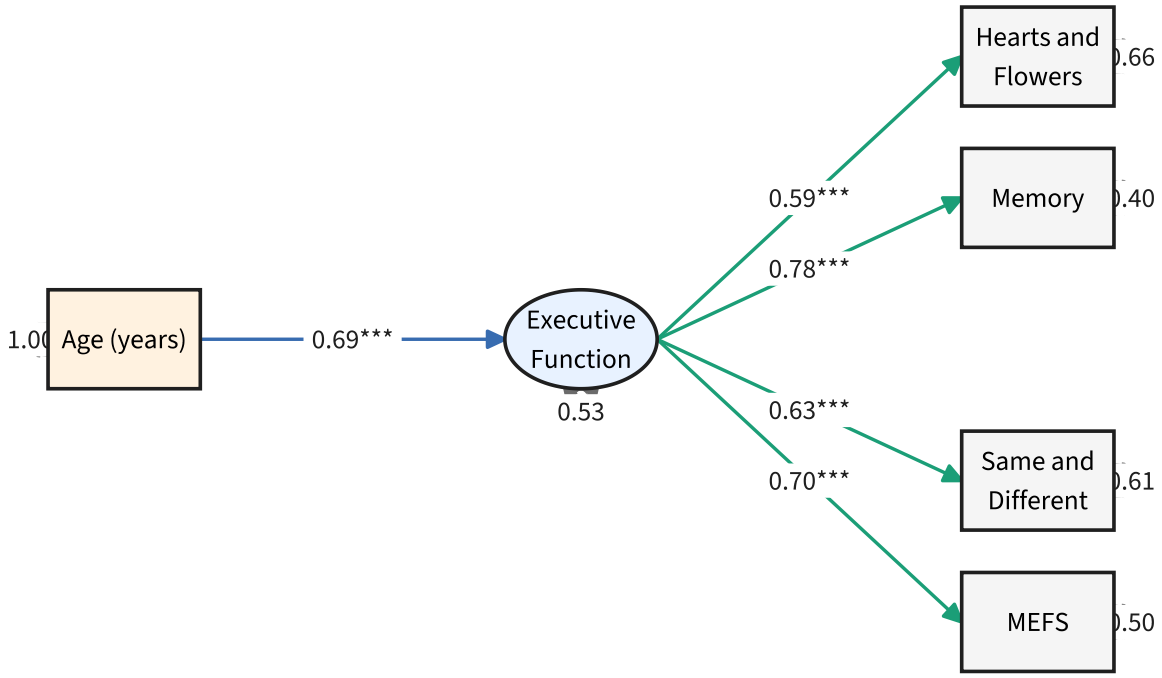
Comparison of alternative structural models including age as a predictor (robust ML, FIML)

Model comparison with age (robust ML, FIML)					
Model	Chi-square	p-value	TLI	CFI	RMSEA
Single factor + age	352.74	< .001	0.86	0.89	0.09
Two factors + age	328.94	< .001	0.86	0.90	0.09
Three factors + age	320.51	< .001	0.85	0.90	0.10

External validity

To assess external validity for the executive function tasks, we fit a single-factor executive function model with Hearts and Flowers, Memory, Same and Different, and the external measure, the Minnesota Executive Function Scale (MEFS) as indicators (Figure 6). We modeled age as a predictor of the latent executive function factor. Model fit did not meet conventional standards, CFI: 0.904, RMSEA: 0.145, RMSEA p-value: <.001. Standardized loadings were moderate-to-strong: Hearts and Flowers = 0.59, Memory = 0.78, Same and Different = 0.63, MEFS = 0.70, indicating that MEFS aligns closely with the latent executive function construct and providing good evidence for external validity. Age strongly predicted executive function (β = 0.69; R^2 = 0.47), consistent with expected developmental gains.

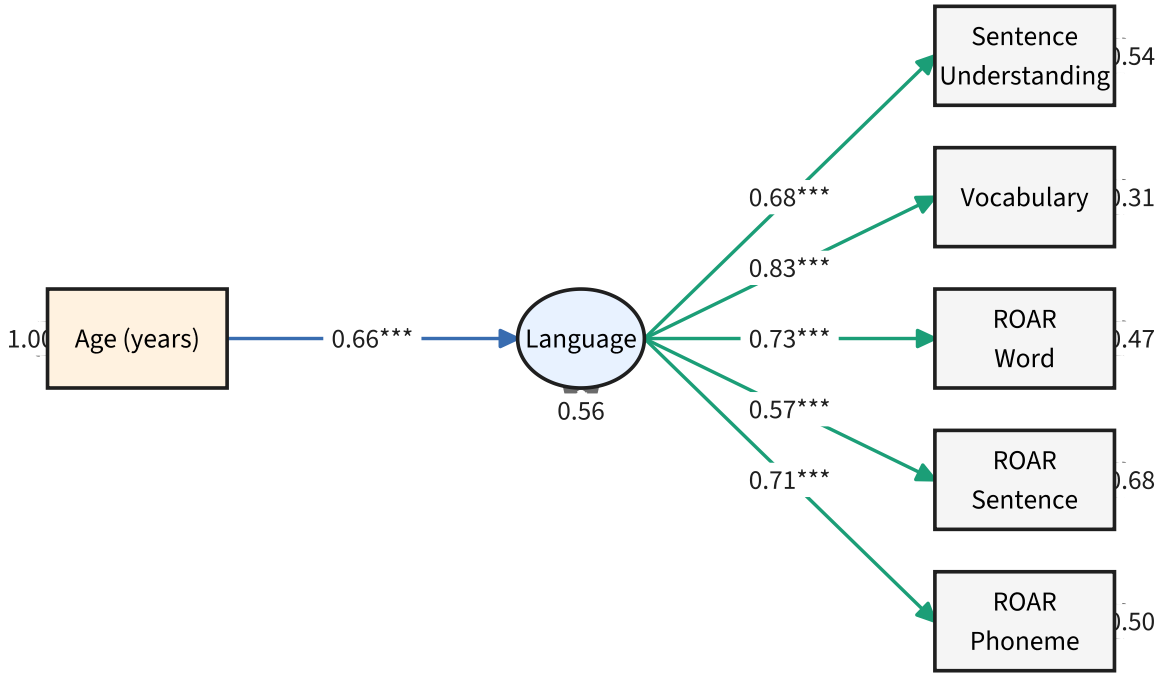
Figure 6
SEM for Executive Function model. Note: Ovals represent the latent factor and rectangles represent observed task scores. Orange denotes the age predictor. Green arrows show factor loadings and the blue arrow shows the regression of the latent factor on age. Values on paths are standardised coefficients.



We ran a similar model to determine the external validity of the language and literacy measures (Sentence Understanding and Vocabulary) against ROAR tasks, including measures of Phonological Awareness (ROAR-Phoneme), Single Word Reading (ROAR-Word) and Sentence Reading Efficiency (ROAR-Sentence). We fit a single latent Language factor with ROAR tasks plus Sentence Understanding and Vocabulary as indicators, and modeled age as a predictor of the latent factor (Figure 7). Model fit again did not meet conventional thresholds, CFI: 0.938, RMSEA: 0.073, RMSEA p-value: .019. Standardized loadings were moderate-to-strong: ROAR-Word = 0.73, ROAR-Sentence = 0.57, ROAR-Phoneme = 0.71, Sentence Understanding = 0.68, Vocabulary = 0.83. Age predicted Language (β = 0.66; R^2 = 0.44) consistent with expected developmental differences.

Figure 7

SEM for Language model. Note: Ovals represent the latent factor and rectangles represent observed task scores. Orange denotes the age predictor. Green arrows show factor loadings and the blue arrow shows the regression of the latent factor on age. Values on paths are standardised coefficients.



Adaptive Task Construction

We have already adapted and piloted several of the LEVANTE core tasks as computer adapted versions, primarily in the mpieva-de and western_ca pilot sites. To date, these include Sentence Understanding, Vocabulary, Shape Rotation, Pattern Matching, Same and Different, and Math. These adaptive versions of the tasks maintain an ability score, θ, as an estimate of the participant's skill level that is updated at the end of each trial. This updated θ is then used to select an item best suited to their estimated ability, which both improves test-taker experience and yields more information on participant skill per item, allowing for a shorter task with fewer items. See Figure 8 for age-related ability trends in the tasks currently using the adaptive format.

The CAT tasks use an adaptive algorithm made available by the open source jsCat JavaScript library (Ma et al., 2025b). The present LEVANTE CAT implementation varies difficulty and guessing for each item while holding both discrimination and upper asymptote constant at 1. Items are selected based on Maximum Fisher Information, and θ is updated according to a maximum likelihood estimator, with limits of -6 and 6. The LEVANTE CATs are configurable with respect to the initial value of theta and the conditions for ending the task. The starting θ is set at 0 for all CAT tasks currently in use, but could in principle be set according to the researcher's prior expectation of participant ability, for example, by using age as a proxy indicator of the initial ability estimate.

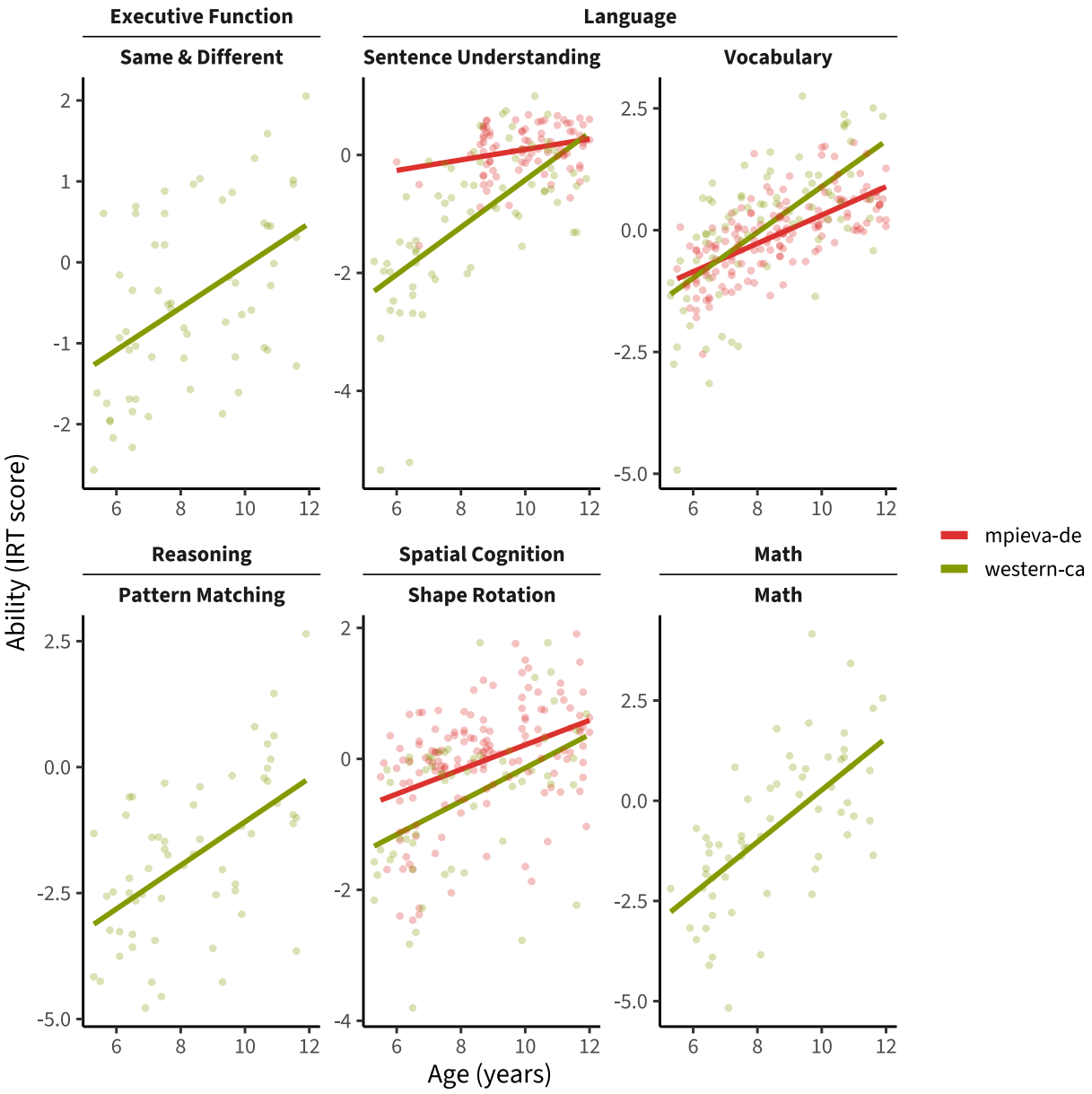
Current CAT implementations use time-based stopping rules to facilitate group testing so that all children end at approximately the same time. Sentence Understanding, Shape Rotation, and Vocabulary each have time limits currently set to 4 minutes. Pattern Matching was set to 6 minutes to allow for the increased time typically required to complete items in this task. Items in these tasks are presented together in a single block.

Same and Different and Math are each divided into three blocks presented sequentially, with per-block stopping based on number of items. These CATs select from the list of items specific to their current block and proceed to the next block once the target number of items is reached, maintaining one θ estimate for the entire task. Same and Different and Math have time

limits of 6 minutes and 8 minutes, respectively.

Figure 8

Age related trends by task for children who completed the adaptive versions.



Standard errors of IRT ability estimates were generally comparable across adaptive and fixed-length versions; typical SE ≈ 0.4–0.8 (see Figure A2 in appendices). In most tasks, the fixed-length forms yielded slightly lower SEs than the adaptive forms across the mid to high range of ability, whereas adaptive forms showed marginally lower SEs for a small number of children at

the lowest ability levels. Given the small cell sizes in these extreme bands and the modest magnitude of differences, precision of ability estimates was broadly similar across administration modes.

General Discussion

In this paper, we presented the LEVANTE core tasks – nine psychometrically grounded tasks for children aged 5–12 that provide efficient, reliable, and valid assessment across language, mathematics, reasoning, executive function, and social cognition. LEVANTE is designed to tackle three core challenges: to span development and enable valid comparisons between younger and older children; to ensure cross-cultural comparability so that measures function equivalently across diverse contexts; and to make the tasks openly available (non-commercial) to maximize access, transparency, and reproducibility.

Current status of the core tasks

Using pilot data from three sites on three continents, collected via three different administration methods (in-school, in-lab, and at-home administration), we provided preliminary evidence of developmental sensitivity, reliability, and validity. Further, using computer adaptive testing algorithms, we are able to deploy efficient versions of these tasks that produce scores for children across a wide range of abilities in just a few minutes. While these tasks do not all yet meet conventional thresholds for reliability, it is important to note that they still fill a gap. Many tasks commonly used to measure these constructs with children have never been assessed for reliability and may well be substantially less reliable than those presented here.

Several of the nine tasks remain under construction, and we expect gains in reliability and validity as items are refined, distractors improved, and instructions localized. We are actively monitoring task performance and iteratively refining items as new data accrue across sites. This includes updating IRT calibrations, conducting routine checks for differential item functioning, and making revisions guided by fit diagnostics and error patterns. We are attempting to strike a balance between minimizing bias and ensuring that observed differences reflect genuine developmental variation rather than artifacts of the measurement process. With pilot data

collection complete, our goal here has been to present the LEVANTE framework tasks, our scoring methods, and an initial analysis of validity and reliability. Although task refinement is an iterative process, we discuss the next steps of extending the framework for broader participation in data collection and analysis.

Future directions

Our plan is for the core tasks presented here to become the backbone of the LEVANTE framework, in which sites around the world collaborate to collect longitudinal data measuring children's learning and development as well as the overlapping contexts in which they develop. De-identified data collected using the tasks presented here flow into a shared repository, becoming part of a large-scale dataset that we hope will enable a wide variety of downstream investigations. The power of these data is that – especially when combined with contextual data about children's caregivers, home environment, school, and community environment – they allow analysis of the sources of developmental and contextual variability at global scale.

Towards this goal, one strength of our approach here is that it easily accommodates future internationalization and adaptation. In fact, at the time of writing we are working on adaptations for two more languages (French and Dutch) and we intend to further broaden the set of languages for which the tasks are available. As part of this process, we have created a website for generating and reviewing translation materials. The LEVANTE researcher website documents the use of this platform and our ongoing language adaptation efforts: https://researcher.levante-network.org.

Our current task set is designed for children ages 5–12, but we are currently pursuing downward extension of many of these tasks for children ages 2–4. By design, the forced choice, tablet-based format of our tasks makes them highly amenable to use with younger children (Frank et al., 2016). In our downward extension work, we are adding easier items to each task, increasing the number of practice trials, and increasing the amount of instructions. We anticipate that not all tasks will be usable for all ages, but for at least the executive function and language tasks there is strong evidence of usability with children under age 5 (Bishop, 1982; Davidson et al., 2006; Long et al., 2025).

Conclusions

In sum, we have taken steps here towards creating a core set of tasks for measuring children's learning and development. We hope that this work is the beginning, rather than the end, of a process of iterative refinement and growth that results in a common, open measurement toolkit for developmental science.

Declarations

Funding

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Conflicts of interest

The authors declare that they have no conflicting interests.

Ethics approval

Each LEVANTE site is responsible for obtaining ethical approval from the relevant institutional and/or governmental ethics committees. Only fully de-identified data are contributed to the LEVANTE Data Repository (LDR).

Consent to participate

Each participating site is responsible for obtaining informed consent in accordance with local regulations and institutional guidelines. Only fully de-identified data enter the LDR, and participant identities cannot be readily ascertained or otherwise associated with the data by LDR staff or secondary data users.

Consent for publication

All authors have consented to the submission of this manuscript for publication. Consent for publication from participants is not applicable because only de-identified, aggregate data are reported.

Availability of data and materials

Data for reproducing this paper are available at https://github.com/levante-framework/levante-pilots.

Code availability

Code for reproducing this paper is available at https://github.com/levante-framework/levante-pilots.

Authors' contribution

Authors' contributions are detailed on the title page, following the CRediT authorship taxonomy.

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Appendix

Age Slopes by Task

Figure A1

Site-specific age slopes by task (EMTrends with 95% CI), non-adaptive data only

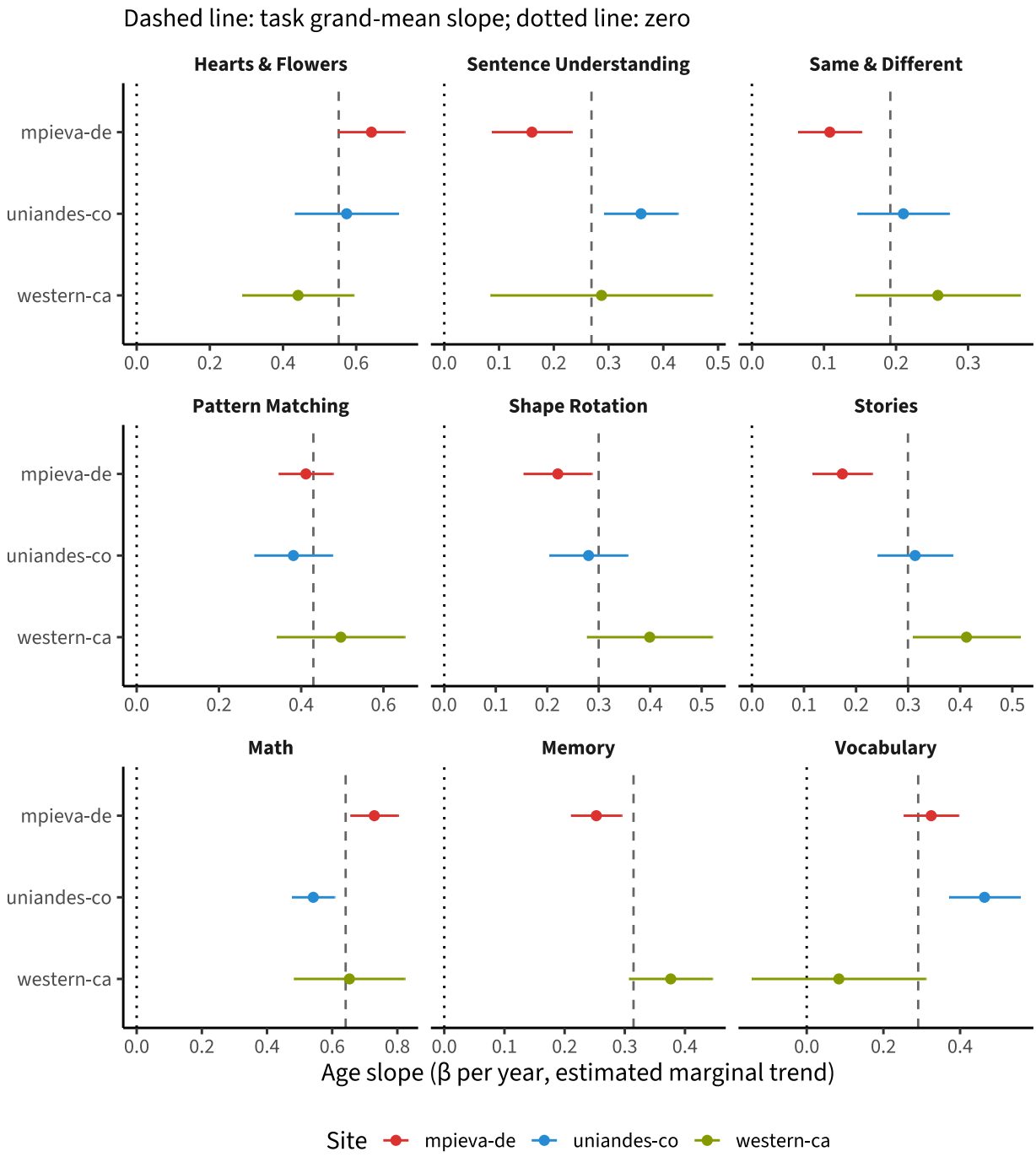
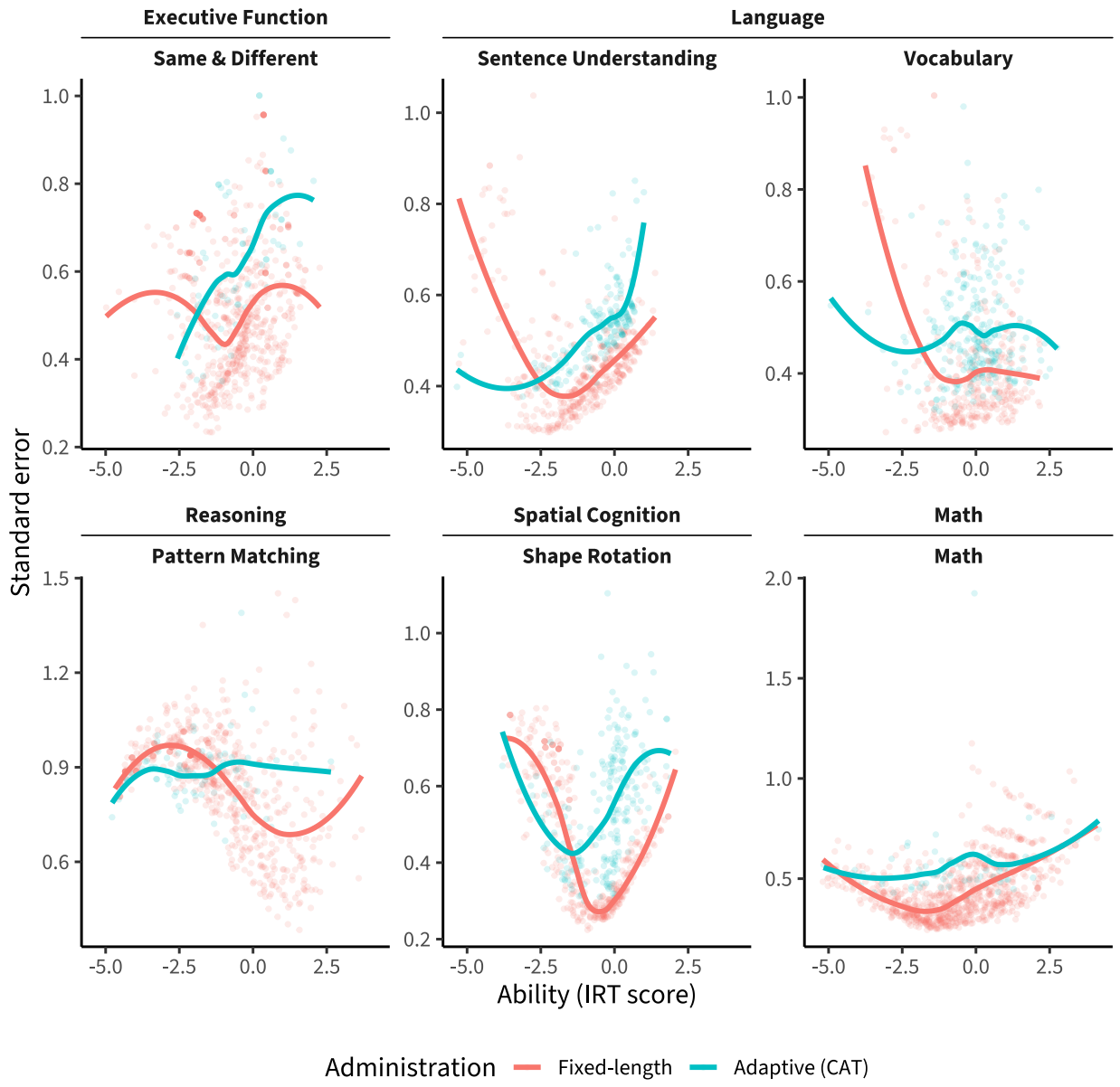


Figure A2

Standard error of IRT ability estimates by ability level for fixed-length versus adaptive versions of the six CAT tasks.